

Decent Coaching Classes / X Maths-I

Solve all the questions. Show all steps.

Section A: Linear Equations in Two Variables

Q1. Solve by elimination method: $3x + 2y = 13$ and $2x - y = 4$.

Q2. Solve using Cramer's rule: $2x + 3y = 13$ and $5x - 2y = 4$.

Q3. The sum of the digits of a two-digit number is 9. The number obtained by reversing the digits is 27 more than the original number. Find the original number.

Section B: Quadratic Equations

Q4. Solve by factorisation method: $2x^2 - 5x - 3 = 0$.

Q5. Solve using the quadratic formula: $x^2 - 4x - 3 = 0$.

Q6. Find the value of k for which the equation $x^2 + kx + 9 = 0$ has real and equal roots.

Section C: Arithmetic Progression

Q7. Find the 20th term of the A.P. 7, 11, 15, ...

Q8. Find the sum of the first 25 terms of the A.P. 3, 8, 13, ...

Q9. In an A.P., the 5th term is 22 and the 9th term is 38. Find the first term, the common difference and the sum of the first 10 terms.

Section D: Financial Planning

Q10. A shopkeeper in Maharashtra sells an article of taxable value ₹12,000 to a customer in the same state. The rate of GST is 12%. Find the CGST, the SGST and the total invoice amount.

Q11. Mr. Patil invested ₹36,000 in shares of face value ₹100 bought at a premium of ₹20. The company declared a dividend of 15%. Find the number of shares he bought, the total dividend he received and his rate of return.

Section E: Probability

Q12. A card is drawn at random from a well-shuffled pack of 52 playing cards. Find the probability that the card is (i) a king (ii) a red card (iii) a face card.

Q13. Two dice are thrown simultaneously. Find the probability that (i) the sum of the numbers is 9 (ii) the sum of the numbers is at least 10.

Section F: Statistics

Q14. Find the mean of the following data by the direct method:

Class	0–10	10–20	20–30	30–40	40–50
Frequency	5	8	12	10	5

Q15. Find the mean of the following data by the assumed mean method (take $A = 45$):

Class	20–30	30–40	40–50	50–60	60–70
Frequency	4	6	8	7	5

Answer Key & Solutions

Q1. Solve by elimination method: $3x + 2y = 13$ and $2x - y = 4$.

$$3x + 2y = 13 \dots (i) \quad 2x - y = 4 \dots (ii)$$

$$\text{Multiply (ii) by 2: } 4x - 2y = 8 \dots (iii)$$

$$\text{Add (i) and (iii): } 7x = 21, \text{ so } x = 3$$

$$\text{Put } x = 3 \text{ in (ii): } 6 - y = 4, \text{ so } y = 2$$

$$\mathbf{x = 3, y = 2}$$

Q2. Solve using Cramer's rule: $2x + 3y = 13$ and $5x - 2y = 4$.

$$D = (2)(-2) - (3)(5) = -4 - 15 = -19$$

$$D_x = (13)(-2) - (3)(4) = -26 - 12 = -38$$

$$D_y = (2)(4) - (13)(5) = 8 - 65 = -57$$

$$x = D_x/D = -38/-19 = 2; \quad y = D_y/D = -57/-19 = 3$$

$$\mathbf{x = 2, y = 3}$$

Q3. The sum of the digits of a two-digit number is 9. The number obtained by reversing the digits is 27 more than the original number. Find the original number.

Let the tens digit be x and the units digit be y . Number = $10x + y$

$$x + y = 9 \dots (i)$$

$$\text{Reversed number} = 10y + x, \text{ so } (10y + x) - (10x + y) = 27, \text{ i.e. } y - x = 3 \dots (ii)$$

$$\text{Add (i) and (ii): } 2y = 12, \text{ so } y = 6 \text{ and } x = 3$$

The number is 36

Q4. Solve by factorisation method: $2x^2 - 5x - 3 = 0$.

$$2x^2 - 5x - 3 = 0$$

$$2x^2 - 6x + x - 3 = 0$$

$$2x(x - 3) + 1(x - 3) = 0$$

$$(x - 3)(2x + 1) = 0$$

$$\mathbf{x = 3 \text{ or } x = -1/2}$$

Q5. Solve using the quadratic formula: $x^2 - 4x - 3 = 0$.

$$a = 1, b = -4, c = -3$$

$$b^2 - 4ac = 16 + 12 = 28$$

$$x = (4 \pm \sqrt{28})/2 = (4 \pm 2\sqrt{7})/2$$

$$\mathbf{x = 2 + \sqrt{7} \text{ or } x = 2 - \sqrt{7}}$$

Q6. Find the value of k for which the equation $x^2 + kx + 9 = 0$ has real and equal roots.

$$\text{For real and equal roots, } b^2 - 4ac = 0$$

$$k^2 - 4(1)(9) = 0, \text{ so } k^2 = 36$$

$$\mathbf{k = 6 \text{ or } k = -6}$$

Q7. Find the 20th term of the A.P. 7, 11, 15, ...

$$a = 7, d = 11 - 7 = 4, n = 20$$

$$t_n = a + (n - 1)d$$

$$t_{20} = 7 + 19 \times 4 = 7 + 76$$

$$\mathbf{t_{20} = 83}$$

Q8. Find the sum of the first 25 terms of the A.P. 3, 8, 13, ...

$$a = 3, d = 5, n = 25$$

$$S_n = n/2 [2a + (n - 1)d]$$

$$S_{25} = 25/2 [6 + 24 \times 5] = 25/2 \times 126$$

$$S_{25} = \mathbf{1575}$$

Q9. In an A.P., the 5th term is 22 and the 9th term is 38. Find the first term, the common difference and the sum of the first 10 terms.

$$t_5 = a + 4d = 22 \dots (i) \quad t_9 = a + 8d = 38 \dots (ii)$$

$$\text{Subtract (i) from (ii): } 4d = 16, \text{ so } d = 4$$

$$\text{From (i): } a + 16 = 22, \text{ so } a = 6$$

$$S_{10} = 10/2 [2(6) + 9(4)] = 5 \times 48$$

$$\mathbf{a = 6, d = 4, S_{10} = 240}$$

Q10. A shopkeeper in Maharashtra sells an article of taxable value ₹12,000 to a customer in the same state. The rate of GST is 12%. Find the CGST, the SGST and the total invoice amount.

$$\text{GST} = 12\% \text{ of } ₹12,000 = ₹1,440$$

$$\text{CGST} = \text{SGST} = 1,440 \div 2 = ₹720 \text{ each}$$

$$\text{Invoice amount} = 12,000 + 1,440$$

$$\mathbf{\text{CGST} = ₹720, \text{SGST} = ₹720, \text{invoice amount} = ₹13,440}$$

Q11. Mr. Patil invested ₹36,000 in shares of face value ₹100 bought at a premium of ₹20. The company declared a dividend of 15%. Find the number of shares he bought, the total dividend he received and his rate of return.

$$\text{Market value of one share} = 100 + 20 = ₹120$$

$$\text{Number of shares} = 36,000 \div 120 = 300$$

$$\text{Dividend per share} = 15\% \text{ of } 100 = ₹15$$

$$\text{Total dividend} = 300 \times 15 = ₹4,500$$

$$\text{Rate of return} = (4,500/36,000) \times 100 = 12.5\%$$

$$\mathbf{300 \text{ shares, dividend} = ₹4,500, \text{rate of return} = 12.5\%}$$

Q12. A card is drawn at random from a well-shuffled pack of 52 playing cards. Find the probability that the card is (i) a king (ii) a red card (iii) a face card.

$$n(S) = 52$$

$$(i) P(\text{king}) = 4/52 = 1/13$$

$$(ii) P(\text{red card}) = 26/52 = 1/2$$

$$(iii) P(\text{face card}) = 12/52 = 3/13$$

$$\mathbf{\text{Answers: } 1/13, 1/2, 3/13}$$

Q13. Two dice are thrown simultaneously. Find the probability that (i) the sum of the numbers is 9 (ii) the sum of the numbers is at least 10.

$$n(S) = 6 \times 6 = 36$$

$$(i) A = \{(3,6), (4,5), (5,4), (6,3)\}, n(A) = 4, \text{ so } P(A) = 4/36 = 1/9$$

$$(ii) B = \{(4,6), (5,5), (6,4), (5,6), (6,5), (6,6)\}, n(B) = 6, \text{ so } P(B) = 6/36 = 1/6$$

$$\mathbf{\text{Answers: } 1/9, 1/6}$$

Q14. Find the mean of the following data by the direct method:

Class marks x_i : 5, 15, 25, 35, 45

$f_i x_i$: 25, 120, 300, 350, 225

$\Sigma f_i = 40$, $\Sigma f_i x_i = 1020$

Mean = $\Sigma f_i x_i / \Sigma f_i = 1020/40$

Mean = 25.5

Q15. Find the mean of the following data by the assumed mean method (take $A = 45$):

Class marks x_i : 25, 35, 45, 55, 65; $A = 45$

$d_i = x_i - A$: -20, -10, 0, 10, 20

$f_i d_i$: -80, -60, 0, 70, 100; $\Sigma f_i d_i = 30$, $\Sigma f_i = 30$

Mean = $A + \Sigma f_i d_i / \Sigma f_i = 45 + 30/30$

Mean = 46